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In [2]: # Program 8: One-sample t-test
import pandas as pd
from scipy import stats

FILE = r"D:\Datascience-MachineLearning\Datasets\diamonds.csv" # <-- change to your file path
COLUMN = None # column name; None = first numeric column
POP_MEAN = 70 # hypothesised population mean

df = pd.read_csv(FILE)
if COLUMN is None:
    COLUMN = df.select_dtypes(include="number").columns[0]
sample = df[COLUMN].dropna()

t, p = stats.ttest_1samp(sample, POP_MEAN)

print("Column      :", COLUMN)
print("Sample mean :", sample.mean())
print("t-statistic :", t)
print("p-value      :", p)

if p < 0.05:
    print("Result: Reject H0 - the mean differs significantly from", POP_MEAN)
else:
    print("Result: Fail to reject H0 - no significant difference from", POP_MEAN)

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Column      : carat
Sample mean : 0.7979397478680014
t-statistic : -33906.72557219791
p-value      : 0.0
Result: Reject H0 - the mean differs significantly from 70

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In [6]: # Program 8: One-sample t-test
import pandas as pd
from scipy import stats

FILE = r"D:\Datascience-MachineLearning\Datasets\diabetes.csv" # <-- change to your file path
COLUMN = None # column name; None = first numeric column
POP_MEAN = 4 # hypothesised population mean

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df = pd.read_csv(FILE)
if COLUMN is None:
    COLUMN = df.select_dtypes(include="number").columns[0]
sample = df[COLUMN].dropna()

t, p = stats.ttest_1samp(sample, POP_MEAN)

print("Column      :", COLUMN)
print("Sample mean :", sample.mean())
print("t-statistic :", t)
print("p-value     :", p)

if p < 0.05:
    print("Result: Reject H0 - the mean differs significantly from", POP_MEAN)
else:
    print("Result: Fail to reject H0 - no significant difference from", POP_MEAN)

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Column      : Pregnancies
Sample mean : 3.8450520833333335
t-statistic : -1.2743561796750649
p-value     : 0.20292317503312896
Result: Fail to reject H0 - no significant difference from 4

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In []: